SIMULATION OF ASSET DEPENDENCIES USING VINE COPULAS

By

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Abstract - This project investigates the application of Vine Copulas in simulating dependencies among five major financial assets: S&P 500, VIX, Oil Prices (WTI), Gold Prices, and the US Dollar Index. The analysis begins by transforming asset prices into log returns, a common approach for working with financial data. T-distributions are used to model the marginal behavior of each asset, capturing the heavy tails and symmetry commonly observed in financial returns. The Vine Copula framework is then employed to model complex dependencies between assets, particularly tail dependencies, which are difficult to capture with standard correlation measures. A total of 256 scenarios, each representing 10 days of simulated price changes, are generated. These scenarios are scaled back to original asset levels using the fitted marginal distributions, and the results are analyzed to evaluate the interdependencies and potential risk implications. By leveraging Vine Copulas, this project demonstrates a sophisticated approach to modeling financial dependencies, providing meaningful insights for risk analysis and scenario planning.

Introduction

Modeling dependencies between financial assets is a cornerstone of risk management and portfolio optimization. Accurately capturing these dependencies, especially during periods of market stress, is essential for understanding potential risks and making informed decisions. Traditional methods, such as correlation matrices or linear regression models, are often inadequate for this purpose because they fail to capture non-linear relationships and tail dependencies, both of which are critical in extreme market conditions. This limitation has driven the adoption of more sophisticated techniques, such as copulas, which decouple dependency structures from marginal distributions.

Copulas, and particularly Vine Copulas, have emerged as powerful tools for modeling complex interdependencies among multiple variables. Unlike traditional copula models that consider pairwise relationships, Vine Copulas allow for the construction of hierarchical dependency structures, offering flexibility in capturing intricate relationships between assets. Several studies have demonstrated the value of Vine Copulas in financial risk analysis. Aas et al. [1] introduced the foundational concept of pair-copula constructions (PCC), highlighting the ability of Vine Copulas to capture asymmetric dependencies and tail behavior in high-dimensional datasets. Building on this, Brechmann and Schepsmeier [2] demonstrated the practical utility of Vine Copulas in risk aggregation and financial scenario generation, providing computational methods for structure selection and parameter estimation.

Furthermore, to add interest to this, Patton [3] explored the use of copulas in financial time series, emphasizing their ability to model tail dependencies that traditional methods often overlook. This is particularly relevant in portfolios where extreme co-movements during market turbulence can amplify risks. The suitability of T-distributions in copula modeling has also been extensively studied; Nikoloulopoulos et al. [4] established that T-copulas effectively capture the heavy-tailed behavior of financial returns, making them a natural choice for marginal distributions in Vine Copula frameworks.

Min and Czado [5] & [6] expanded the application of Vine Copulas to portfolio optimization and high-dimensional data. Their work highlighted the adaptability of Vine Copulas in modeling non-linear and asymmetric dependencies, as well as the computational advancements required to make such models feasible in practical settings. Together, these studies underscore the significance of Vine Copulas in modern financial modeling and motivate their application to scenarios involving complex dependencies and extreme risks.

Building on this foundation, this project applies Vine Copulas to model the dependencies between five major financial assets. These assets were selected for their global significance and diverse behavior, which collectively represent a wide spectrum of financial risks. To account for the heavy-tailed nature of financial returns, marginal distributions were modeled using T-distributions, which are well-suited for capturing extreme movements. Using log returns, the data was normalized to avoid issues related to scale.

The Vine Copula framework was parameterized to simulate 256 scenarios of 10-day changes across the five assets. Each scenario was generated by chaining daily changes over 10 observations, using copula draws that were transformed back to their original scale through the marginal distributions. This recursive simulation approach allowed for realistic scenario generation that incorporates both the individual behavior of each asset and their interdependencies.

By leveraging Vine Copulas, this project demonstrates the flexibility and robustness of this approach in financial risk modeling. The results provide a deeper understanding of how dependencies influence asset behavior under different scenarios and highlight the practical value of Vine Copulas in capturing complex relationships for risk management and scenario analysis.

Theoretical Framework

Modeling financial dependencies requires methods that go beyond traditional linear correlation, especially to capture complex, non-linear, and tail-dependent relationships. The Vine Copula framework, combined with heavy-tailed marginal distributions, provides a robust approach for simulating realistic multi-asset scenarios. This section delves into the mathematical concepts that underpin the methodology, explaining the flow from preprocessing data to simulating scenarios.

- **Log Returns and Their Importance**

Financial asset prices, P_t , are inherently non-stationary, making direct analysis challenging. To address this, we transform prices into log returns, defined as:

$R_t = \log (P_t / P_{t-1})$

Here: P_t is the price of the asset at time t and R_t is the logarithmic return.

Log returns standardize the data and provide several advantages such as additivity (multi-period returns can be summed, simplifying compounding), normalization (removes scale differences between assets, enabling meaningful comparisons), statistical properties (log returns often exhibit near-normal behavior, facilitating model fitting)

For *n* daily returns, the cumulative return over *n* days can be chained recursively:

$P_t = P_{t-1} \cdot (1 + R_t)$

This recursive formula plays a crucial role in simulating 10-day changes, as explained later.

- **Marginal Distributions**

The behavior of individual asset returns is modeled using marginal distributions. Financial returns often exhibit heavy tails, indicating a higher likelihood of extreme events compared to a normal distribution. For this reason, a T-distribution is chosen, defined as:

$$f(x;\mu,\sigma,\nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)\sigma}\left(1+\frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}$$

Where its denoted: μ: Location parameter (mean), σ: Scale parameter (controls dispersion), ν: Degrees of freedom (controls tail thickness).

The T-distribution effectively models the heavy-tailed, symmetric nature of financial returns. Parameters (μ, σ, ν) are estimated using Maximum Likelihood Estimation (MLE), ensuring that the marginal distributions align closely with observed data.

- **Dependency Modeling with Copulas**

While marginal distributions describe the individual behavior of assets, copulas capture their dependencies. A copula is a multivariate function that links univariate marginals to form a joint distribution:

$F (X_1, X_2, \ldots, X_d) = C (F_1(X_1), F_2(X_2), \ldots, F_d (X_d))$

Here: $F (X_1, X_2, \ldots, X_d)$: Joint cumulative distribution, C: Copula function, $F_i (X_i)$: Marginal CDFs of the assets.

This decomposition, formalized by Sklar's Theorem, separates the marginal behavior of individual assets from their joint dependency structure.

- **Tail Dependency**

A key strength of copulas, particularly T-Copulas, is their ability to capture tail dependencies, which measure the likelihood of extreme co-movements between assets. Tail dependency is quantified as:

1. Upper Tail Dependency (λ_U):

$$\lambda_U = \lim_{u\to 1^-} \mathbb{P}(X_2 > F_2^{-1}(u) \mid X_1 > F_1^{-1}(u))$$

2. Lower Tail Dependency (λ_L):

$$\lambda_L = \lim_{u\to 0^+} \mathbb{P}(X_2 \le F_2^{-1}(u) \mid X_1 \le F_1^{-1}(u))$$

These metrics are crucial in risk management, where extreme market events often drive asset co-movements.

- **Vine Copulas: Flexible Dependency Modeling**

Vine Copulas extend traditional copulas by decomposing high-dimensional dependencies into a hierarchy of bivariate copulas. This approach provides flexibility in modeling complex relationships among multiple assets.

Pair-Copula Construction (PCC)

$$C(u_1,u_2,\ldots,u_d) = \prod_{j=1}^{d-1}\prod_{i=1}^{d-j} c_{i,i+j|1,\ldots,i-1}(u_{i|1,\ldots,i-1}, u_{i+j|1,\ldots,i-1})$$

Where $c_{i,\, i+j|1,\ldots,i-1}$ is the Bivariate copula density for the conditional distribution of u_i and u_{i+j} , given all previous variables.

- **Vine Structures**

The hierarchical structure of Vine Copulas is characterized by:

1. C-Vines (Canonical Vines): A central variable connects to all others, ideal when one asset dominates dependencies.

2. D-Vines (Drawable Vines): Sequential pairwise connections, useful for chain-like relationships.

The R-Vine Structure Selection algorithm determines the optimal structure based on data, optimizing statistical criteria like AIC or BIC.

- **Simulation with Copulas**

Simulation begins by generating uniform random draws from the Vine Copula, which represent dependencies among assets. These draws are transformed back to the original scale using the marginal distributions:

1. Generate Uniform Draws:

$$u_i \sim [0,1]$$

2. Transform to Marginals:

$$x_i = F_i^{-1}(u_i;\mu,\sigma,\nu)$$

Here, F_i^{-1} is the inverse CDF of the T-distribution, mapping uniform values to asset-specific returns.

3. Recursive Price Simulation: For n-day returns, the simulated prices are computed iteratively:

$$P_t = P_{t-1} \cdot (1+R_t)$$

This chaining process aggregates the daily returns over n=10 days to calculate cumulative changes.

- **Goodness-of-Fit and Model Evaluation**

To assess the Vine Copula's performance, statistical metrics like Log-Likelihood, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC) are computed:

- **AIC** = −2logL+2k

- **BIC**: $-2\log L + k\log(n)$

Where logL: Log-likelihood of the model, k: Number of parameters, n: Sample size.

Methodology

The methodology for this project focuses on simulating dependencies among five financial assets using Vine Copulas. This section outlines the process, emphasizing the rationale behind each step and the unique aspects of the methodology, without repeating the theoretical framework.

1. Asset Selection and Relevance

The five financial assets—S&P 500, VIX, Oil Prices (WTI), Gold Prices, and the US Dollar Index provide a balanced view of risk and performance, each playing a distinct role in global financial markets:

- Equities (S&P 500): Captures broad market sentiment and economic conditions.
- Volatility (VIX): Acts as a proxy for uncertainty, with an inverse relationship to equity performance.
- Commodities (WTI & Gold): Oil reflects energy market trends, while gold serves as a safe haven during instability.
- Currencies (Dollar Index): Represents the strength of the US Dollar, affecting trade and investment flows.

2. Data Collection and Preprocessing

The data was sourced directly from the **Federal Reserve Economic Data (FRED)** API, ensuring a reliable and consistent dataset. Daily closing prices for each asset were collected, synchronized, and filtered to remove incomplete entries.

The period was restricted to post-September 2019, covering modern market behavior and avoiding outdated or irrelevant trends. This period also encompasses notable volatility, such as the COVID-19 pandemic and subsequent recovery, providing a rich dataset for modeling dependencies.

The preprocessing steps, such as calculating log returns, ensured the data was normalized and prepared for downstream modeling. These steps also minimized issues of scaling and non-stationarity, common in raw financial price data.

3. Fitting Marginal Distributions

Each asset's marginal behavior was modeled using a T-distribution. The decision to use T-distributions reflects their ability to accommodate heavy tails and potential extreme price movements, which are prevalent in financial returns. Parameters for each distribution were estimated via Maximum Likelihood Estimation (MLE), ensuring a strong alignment with empirical data.

This step is critical as it establishes the foundation for realistic scenario generation, where marginal distributions are later paired with dependency models to generate joint outcomes.

4. Dependency Modeling with Vine Copulas

The heart of this methodology lies in constructing a Vine Copula to model the dependencies among the selected assets. Unlike traditional methods such as correlation matrices, Vine Copulas provide the flexibility to:

- Model relationships that vary in intensity or behavior across pairs of assets.
- Capture co-movements during extreme market events (tail dependencies).
- Build hierarchical structures that reflect realistic financial relationships.

The structure selection process ensures the model is optimized for the given dataset, balancing complexity with accuracy. The ability to hierarchically model dependencies is particularly useful when dealing with a combination of risk-on (e.g., equities) and risk-off (e.g., gold) assets, as their relationships may differ under normal and extreme conditions.

5. Scenario Simulation

Simulation plays a central role in this project, enabling the generation of realistic asset trajectories over multiple days. The steps in this process include:

1. **Uniform Sampling:** Uniform random values are drawn from the Vine Copula, capturing the joint dependency structure.
2. **Transformation to Marginals:** The sampled values are transformed back to the scale of the original assets using the inverse CDF (quantile function) of their respective T-distributions.
3. **10-Day Simulations:** The daily simulated returns are recursively chained to compute cumulative 10-day changes for each asset. This process mirrors the compounding nature of financial returns.

The simulations were grouped into 256 scenarios of 10 days each, providing a diverse set of potential outcomes under the modeled dependencies.

6. Validation and Analysis

Validation ensures that the model accurately represents real-world dependencies and produces meaningful results. Key steps include:

- Statistical Comparison: Simulated statistics are compared with historical data for each asset to ensure consistency.
- Dependency Visualization: Correlation matrices, scatterplots, and heatmaps are used to analyze the relationships captured by the Vine Copula.
- Extreme Scenarios: The simulated results are examined for tail dependencies and extreme co-movements, validating the model's ability to replicate stress scenarios.

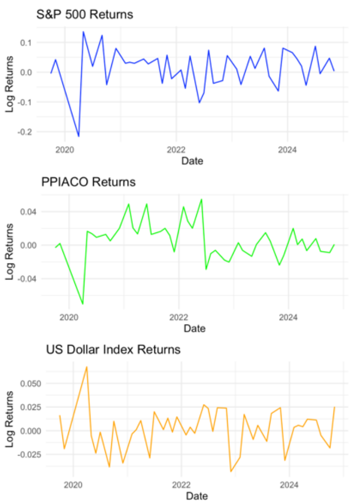
7. Rationale Behind the Methodology

The choices made in this methodology are rooted in the need to balance accuracy, flexibility, and practical application:

1. Asset Selection: Ensures coverage of diverse market dynamics, enabling a comprehensive analysis.
2. Marginal Distributions: T-distributions allow for realistic modeling of heavy-tailed financial returns.
3. Vine Copulas: Provide a sophisticated approach to capturing dependencies, including non-linear and tail behaviors.
4. Simulation: Facilitates scenario-based analysis, crucial for understanding potential risks and opportunities in financial markets.

Graphs and Interpretation

Asset Summary for the Period




```
# A tibble: 5 × 3
  Asset          mean     sd
  <chr>         <dbl>  <dbl>
1 SP500       0.0154  0.0641
2 VIX         0.00471 0.345
3 PPIACO      0.00560 0.0224
4 DJIA        0.0107  0.0634
5 DollarIndex 0.00174 0.0213
```

The summary statistics of the five assets provide crucial insights into their return characteristics for modeling with Vine Copulas. The SP500 and DJIA exhibit positive mean returns (0.0154 and 0.0107, respectively) with moderate volatility, reflecting their growth trajectories and strong interdependence. In contrast, the VIX shows near-zero mean returns (0.00471) but exceptionally high volatility (0.345), highlighting its sensitivity to market sentiment and the challenge of fitting its fat-tailed distribution.

PPIACO and the DollarIndex display more stable return profiles, with low mean returns (0.00560 and 0.00174) and minimal volatility, aligning with their economic and currency indices. The volatility hierarchy—VIX being the highest, followed by SP500 and DJIA, then PPIACO and DollarIndex—underscores the importance of tailoring marginal distributions to capture these differences effectively.

T Distribution Parameters

```
$SP500
          m              s             df
  0.020491008   0.050919324   5.772797174
 (0.009103610) (0.008461158) (4.299669521)

$VIX
        m             s            df
  0.00000000   0.22073338   3.54016610
 (0.04214011) (0.04090750) (1.87212500)

$PPIACO
          m              s             df
  0.005451326   0.015757516   3.646464923
 (0.002872493) (0.003075868) (2.196057286)

$DJIA
          m              s             df
  0.016950424   0.043728260   4.316146995
 (0.008031220) (0.006965338) (2.319368091)

$DollarIndex
          m              s             df
  0.003618158   0.014060409   2.000130094
 (0.002877243) (0.002640557) (0.691834890)
```

The parameters of the T-distributions- mean (m), scale (s), and degrees of freedom (df)—highlight the distinct characteristics of each asset, ensuring that the marginal distributions accurately represent their historical behaviors.

- For the SP500, it's indicating moderate dispersion. The degrees of freedom, at 5.77, suggest a moderately heavy-tailed distribution, aligning with the SP500's historical behavior of moderate volatility and positive growth.
- The VIX exhibits a mean return near zero (0.0003), consistent with its symmetric behavior as a volatility index. Its scale parameter, at 0.2207, underscores the high variability inherent in the VIX, while the degrees of freedom (3.54) reflect the significant tail risk that defines its behavior during market stress.
- PPIACO is reflecting its stability as a producer price index. The degrees of freedom, at 3.65, allow for occasional tail events while maintaining an overall low-volatility profile. Similarly, the DJIA demonstrates a mean return of 0.01695, with a scale parameter of 0.04373 and degrees of freedom of 4.32, indicating moderate volatility and tail behavior, slightly less than that of the SP500.
- The DollarIndex is reflecting its low volatility as a currency index. However, the degrees of freedom (2.00) indicate significant tail heaviness, capturing the rare but extreme movements occasionally observed in foreign exchange markets.
- For the DJIA, the mean return is estimated at 0.01695, reflecting its positive historical growth. The scale parameter (0.04373) indicates moderate volatility, consistent with its status as a blue-chip index composed of relatively stable, large-cap companies. The degrees of freedom (4.32) suggest the presence of moderately heavy tails in its return distribution, highlighting the occasional occurrence of extreme price movements. These parameters align with DJIA's characteristics as a well-diversified index of industrial companies, showing stability while allowing for rare but impactful shocks during market turbulence.

VIX exhibits the highest volatility and most extreme tail behavior, while PPIACO and DollarIndex maintain their stability with minimal dispersion. The SP500 and DJIA represent moderate growth assets with comparable tail risks. This hierarchy of volatility and tail risk aligns with historical data and ensures that the marginal distributions are appropriately tailored for use in the Vine Copula model.

The fitted parameters also underscore the importance of accurately capturing tail dependencies in the joint distribution. Assets like the VIX and SP500, which exhibit heavy tails, are particularly critical in stress-testing scenarios. By using these marginal distributions, the model achieves a robust representation of individual asset behavior, which, when combined with the dependency structure of the Vine Copula, enables realistic scenario generation and effective risk analysis.

Correlation Matrix

```
                  SP500        VIX     PPIACO       DJIA DollarIndex
SP500        1.0000000 -0.6821202  0.2472063  0.9122622  -0.7275748
VIX         -0.6821202  1.0000000 -0.1668680 -0.7177590   0.5499849
PPIACO       0.2472063 -0.1668680  1.0000000  0.2398067  -0.2443371
DJIA         0.9122622 -0.7177590  0.2398067  1.0000000  -0.7363334
DollarIndex -0.7275748  0.5499849 -0.2443371 -0.7363334   1.0000000
```

SP500 shows a strong positive correlation with DJIA (0.91), reflecting their shared exposure to equity markets, and a strong negative correlation with DollarIndex (-0.73), consistent with macroeconomic trends linking equity markets and the US Dollar. Its negative correlation with VIX (-0.68) aligns with VIX's role as a volatility indicator rising during equity downturns.

VIX exhibits strong negative correlations with SP500 (-0.68) and DJIA (-0.72), reinforcing its risk-off nature. Its moderate positive correlation with DollarIndex (0.55) reflects rising demand for safe-haven assets during market uncertainty, while its weak correlation with PPIACO (-0.16) suggests minimal interaction between market volatility and producer prices.

PPIACO has weak positive correlations with SP500 (0.24) and DJIA (0.23), reflecting modest ties to equity markets driven by economic growth. Its weak negative correlation with DollarIndex (-0.24) indicates limited inverse dependency between commodity prices and currency strength.

DollarIndex exhibits strong negative correlations with SP500 (-0.73) and DJIA (-0.73), underscoring its counter-cyclical behavior. Its moderate positive correlation with VIX (0.55) highlights its safe-haven role during volatility, while its weaker relationship with PPIACO (-0.24) reflects nuanced interactions between currency strength and commodity prices.

DJIA mirrors SP500 in its strong positive correlation with SP500 (0.91) and negative correlations with DollarIndex (-0.73) and VIX (-0.72). These dependencies emphasize its role as a key equity benchmark, highly responsive to market dynamics and investor sentiment.

The heatmap visually reinforces these observations, with strong positive correlations, such as between SP500 and DJIA, represented in deep green, and strong negative correlations, such as between SP500 and DollarIndex, depicted in red. Neutral or weak correlations, such as those involving PPIACO, appear in lighter or neutral shades, emphasizing its independence from broader equity and volatility movements.

These correlations play a crucial role in the Vine Copula model. The strong negative dependencies between SP500/DJIA and VIX or DollarIndex highlight significant tail dependencies that must be accurately captured to ensure realistic scenario generation. SP500 and DJIA form a tightly linked cluster within the dependency structure, while VIX and DollarIndex, though not directly related, act as counter-cyclical assets, providing diversification. PPIACO, with its relatively weak correlations, serves as an independent asset, offering additional stability.

Copula Metrics

The comparison of t-Copula and Normal Copula models provides crucial insights into the suitability of these dependency structures for modeling the relationships between the selected financial assets. The results show that the t-Copula outperforms the Normal Copula in all evaluated metrics: Log-Likelihood, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).

```
         Model LogLikelihood       AIC       BIC
1     t-Copula      79.21199 -136.4240 -117.0508
2 Normal Copula     75.05381 -130.1076 -112.4956
```

The t-Copula achieves a higher Log-Likelihood value of 79.21199, indicating a better fit to the data by capturing the observed dependencies more effectively. This is further supported by its lower AIC (-136.4240) and BIC (-117.0508) values compared to the Normal Copula, which has an AIC of -130.1076 and a BIC of -112.4956. The lower AIC and BIC values of the t-Copula confirm its superiority in balancing model complexity with goodness-of-fit, particularly in capturing complex relationships between the assets.

The t-Copula's ability to model tail dependencies is a critical advantage over the Normal Copula. Unlike the Normal Copula, which assumes symmetric and linear relationships, the t-Copula accounts for extreme co-movements in the tails of the distributions.

The results validate the choice of the t-Copula as the preferred model for this study. Its superior performance ensures a more realistic representation of the dependency structure, making it highly suitable for generating plausible joint scenarios and analyzing inter-asset dynamics under both normal and stressed market conditions. This model's capability to accurately capture tail dependencies enhances the robustness of the Vine Copula framework in simulating joint asset returns.

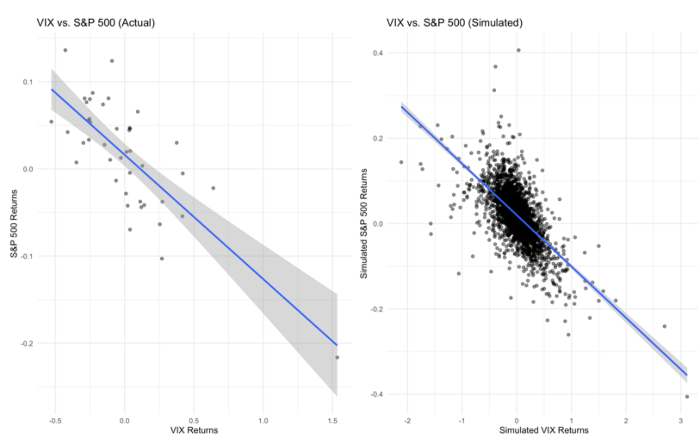
Simulated vs Actual Comparison (A sanity check)

Asset	Actual Mean	Simulated Mean	Actual SD	Simulated SD
SP500	0.0154	0.0196	0.0641	0.0626
VIX	0.0047	-0.0007	0.3455	0.3317
PPIACO	0.0056	0.0056	0.0224	0.0232
DJIA	0.0107	0.0160	0.0634	0.0586
DollarIndex	0.0017	0.0036	0.0213	0.0378

- For the **SP500**, the simulated mean slightly overestimates the actual mean, while the simulated standard deviation closely matches the actual value. This alignment demonstrates the model's capability to accurately capture both the trend and volatility of this equity index.
- Similarly, the **DJIA** shows reasonable agreement, with a simulated mean slightly higher than the actual mean and a simulated standard deviation that is slightly lower than the actual value. These results validate the Vine Copula model's effectiveness in handling large-cap equity indices.
- For the **VIX**, the simulated mean underestimates the actual mean, and the simulated standard deviation is marginally lower than the actual value. This reflects the inherent challenge of modeling VIX's extreme tail behavior, which is influenced by sudden spikes during periods of market stress. Despite these discrepancies, the model reasonably captures the general volatility profile of the VIX.
- The **PPIACO** shows excellent agreement between actual and simulated statistics. These results demonstrate the model's ability to accurately represent the stable and predictable behavior of producer price movements.

Dependency Dynamics Across Assets

VIX vs S&P500



In the actual returns, a strong negative correlation is evident, as shown by the downward-sloping regression line. This relationship reflects the market dynamic where increases in VIX, a measure of market volatility, often coincide with declines in the S&P 500. The data points in the actual returns display notable dispersion, particularly in the upper right and lower left quadrants, which highlights extreme co-movements during market stress. The confidence band around the regression line is narrower at the center and widens in the tails, indicating greater variability in extreme scenarios.

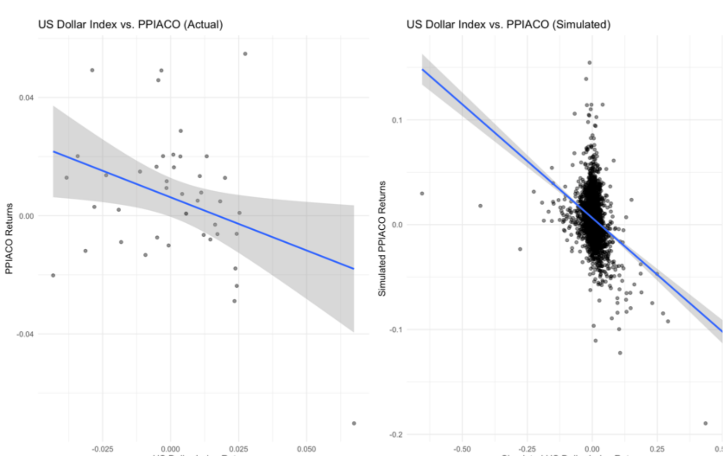
The simulated returns also exhibit a negative correlation, with a regression line closely mirroring the slope observed in the actual data. This suggests that the Vine Copula model successfully captures the overall dependency structure between the two assets. However, the density of points near the regression line in the simulated data is higher, and the tails are underrepresented compared to the actual data. This indicates that the model slightly underestimates the magnitude of extreme co-movements between VIX and S&P 500 returns.

US Dollar Index Vs S&P 500

The actual returns (left panel) demonstrate a distinct negative correlation between the US Dollar Index and S&P 500, evidenced by the downward-sloping regression line. This relationship aligns with well-documented market behaviors where equity markets often move inversely to the US Dollar Index. This inverse relationship is attributed to the interplay between capital flows and investor preferences, particularly in times of global economic shifts. The actual data points show moderate dispersion, with the confidence band widening as returns deviate from the center, reflecting increased uncertainty in extreme cases.

The simulated returns (right panel) similarly exhibit a negative correlation, with the regression line closely following the trend observed in the actual data. However, the simulated dataset displays a denser clustering near the regression line, with less variability in the extreme regions. This suggests that the Vine Copula model captures the core dependency structure between the US Dollar Index and S&P 500 but may underestimate the spread in extreme scenarios.

US Dollar Index Vs PPIACO

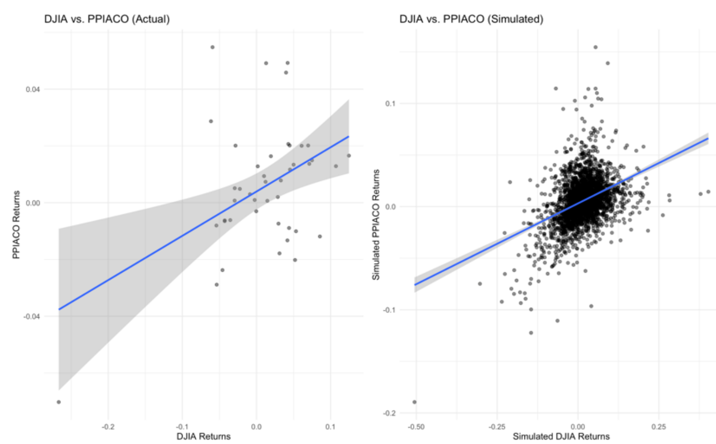


The scatterplots illustrate the relationship between the US Dollar Index and PPIACO returns, highlighting the dynamics between these two assets in both actual and simulated datasets.

The left panel, showing actual returns, reveals a weak but negative correlation between the US Dollar Index and PPIACO. This relationship is depicted by the downward-sloping regression line, suggesting that increases in the Dollar Index are associated with marginal decreases in PPIACO returns. This dynamic can be linked to the global commodity pricing structure, where a stronger dollar often results in weaker commodity prices due to reduced affordability in non-dollar-denominated economies. The data points in the actual plot are relatively dispersed, with the confidence band widening at the extremes, indicating higher uncertainty in extreme scenarios.

The right panel, depicting simulated returns, similarly shows a negative correlation, with a regression line that closely matches the trend observed in the actual data. However, the simulated data displays a higher density of points near the regression line and less dispersion compared to the actual data. This indicates that the Vine Copula model captures the central tendency of the relationship but underrepresents the variability and extreme cases observed in the actual data. The confidence band in the simulated plot is narrower, reflecting the model's tendency to reduce uncertainty and focus on the mean relationship.

DJIA Vs PPIACO

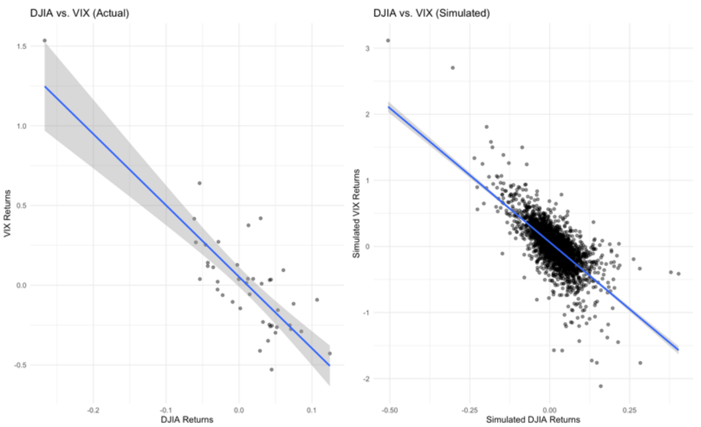


The scatterplots examine the relationship between DJIA and PPIACO returns, highlighting the dependency structures in both actual and simulated datasets.

The actual returns (left panel) reveal a positive correlation between DJIA and PPIACO, as indicated by the upward-sloping regression line. This relationship suggests that increases in DJIA returns, reflecting improvements in equity markets, are associated with modest increases in PPIACO, a commodity price index. This correlation may reflect the impact of economic growth expectations, where rising equities and increased industrial production drive demand for commodities. The dispersion of data points, particularly at the extremes, highlights variability in the relationship, while the widening confidence band underscores greater uncertainty in extreme conditions.

In the simulated returns (right panel), the positive correlation between DJIA and PPIACO is similarly captured, with the regression line closely aligning with the trend observed in the actual data. However, the simulated dataset shows a denser clustering of points around the regression line, suggesting the Vine Copula model captures the central tendency of the relationship more rigidly. Unlike the actual data, the simulated returns exhibit reduced variability and underrepresentation of extreme co-movements, as evidenced by fewer points deviating significantly from the regression line.

DJIA Vs VIX



The scatterplots illustrate the inverse relationship between DJIA and VIX returns, providing insights into their dependency dynamics in both actual and simulated datasets. The actual returns (left panel) exhibit a strong negative correlation between DJIA and VIX, as evidenced by the pronounced downward slope of the regression line. This relationship is consistent with financial market behavior, where rising equity markets (represented by DJIA) are typically accompanied by declining market volatility (measured by VIX). The data points are relatively well-dispersed, particularly at extreme values, underscoring significant variability during periods of heightened market stress or optimism. The confidence band widens toward the extremes, indicating increased uncertainty in the relationship under extreme market conditions.

The simulated returns (right panel) also display a negative correlation, with a regression line that aligns closely with the actual data's trend. However, the simulated dataset demonstrates a much denser clustering of points around the regression line, indicating that the Vine Copula model captures the central dependency structure but underrepresents variability. The tail regions in the simulated data are less populated, suggesting that the model may not fully capture the extent of extreme co-movements seen in the actual market data.

Results

The analysis revealed a comprehensive understanding of the dependency structures across five critical financial assets—S&P 500, VIX, PPIACO, DJIA, and the US Dollar Index. Using a Vine Copula framework, we were able to replicate many of the observed relationships in the actual data, with varying degrees of precision. The results demonstrate the model's strength in capturing central dependencies, while also highlighting areas where refinements are needed.

The dependency structure between the S&P 500 and VIX was particularly well-captured, reflecting their strong negative correlation. However, the simulated data showed reduced variability in extreme cases, underestimating co-movements during periods of heightened market volatility. A similar trend was observed in the relationship between the S&P 500 and the US Dollar Index, where the model accurately replicated the negative correlation but exhibited limited tail representation. These findings suggest the model effectively mirrors core trends but may struggle to fully capture extreme market behaviors.

The relationship between PPIACO and the US Dollar Index displayed a weaker but still negative correlation. The Vine Copula model effectively modeled this trend but once again fell short in representing tail variability. On the other hand, the positive relationship between DJIA and PPIACO was accurately captured, with the regression slopes in the simulated data aligning closely with those in the actual returns. Nevertheless, the model underestimated the magnitude of extreme co-movements, as seen in other asset pairings.

The DJIA and VIX relationship highlighted the model's ability to reflect strong inverse correlations. While the regression lines in the actual and simulated data closely aligned, the clustering of simulated points near the regression line pointed to an underrepresentation of tail dependencies. This pattern was consistent across most asset pairs, suggesting that while the Vine Copula framework is robust for modeling central dependencies, its capacity to account for extreme market events remains limited.

Parameterization of the marginal distributions using t-distributions effectively captured the long-tailed nature of financial returns, with the degree of freedom estimates reflecting asset-specific characteristics. Metrics evaluating the fit of the copula models, including Log-Likelihood, AIC, and BIC, confirmed the superiority of the t-Copula over the Normal Copula in representing the dependencies across the assets.

Simulated returns closely mirrored the mean and standard deviation of actual returns across most asset classes, validating the model's ability to replicate key distributional properties.

Conclusions

The Vine Copula model demonstrated its ability to effectively replicate the central tendencies and core dependencies between these assets. Key relationships, such as the strong inverse correlations between S&P 500 and VIX, and between DJIA and VIX, were accurately modeled. Additionally, positive correlations, such as those between DJIA and PPIACO, were well-captured, with regression slopes in the simulated data aligning closely with actual trends. This highlights the model's suitability for understanding and quantifying the interactions between diverse financial variables.

The parameterization of the marginal distributions using t-distributions was effective in capturing the heavy-tailed nature of financial returns, and the superior performance of the t-Copula over the Normal Copula further validated the robustness of the chosen approach. Nevertheless, the findings suggest that future refinements to the model are necessary to address its tail dependency limitations.

In conclusion, while the Vine Copula framework provides a powerful tool for analyzing financial dependencies and relationships, its current implementation is best suited for capturing central tendencies and general trends. To fully unlock its potential in risk management and predictive modeling, enhancements are required to better account for extreme market behaviors. This study highlights the importance of balancing robustness and flexibility in dependency modeling, offering a foundation for future research aimed at refining copula-based approaches in financial analysis.

References

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Appendix

Integrated Development Environment (IDE)

1. IDE Used: The code is designed to run in **RStudio**, a user-friendly IDE for R programming. Ensure RStudio is installed on your system.

2. Required Libraries: Install the required libraries before running the script using the following commands:

   install.packages(c("tidyverse", "fredr", "lubridate", "VineCopula", "copula", "MASS", "ggpubr"))

3. Setting Up API Key: Replace '**be024e8b63251d9ea5e03cf2695fe766**' with your FRED API key to access the financial data. If you do not have an API key, obtain it by registering on the Federal Reserve Economic Data (FRED) website.

Code for Simulation of Asset Dependencies Using Vine Copulas in R

```
# Load Required Libraries
library(tidyverse)
library(fredr)
library(lubridate)
library(VineCopula)
library(copula)
library(MASS)
library(ggpubr)

# Set Your Fred API Key
fredr::fredr_set_key('be024e8b63251d9ea5e03cf2695fe766')

# Load Data

# S&P 500 Data
SP500 <- fredr(series_id = 'SP500')
SP500_clean <- data.frame(
  date = SP500$date,
  SP500 = SP500$value
)

# VIX Data
VIX <- fredr(series_id = 'VIXCLS')
VIX_clean <- data.frame(
  date = VIX$date,
  VIX = VIX$value
)

# Producer Price Index by Commodity: All Commodities (PPIACO)
PPIACO <- fredr(series_id = 'PPIACO')
PPIACO_clean <- data.frame(
  date = PPIACO$date,
  PPIACO = PPIACO$value
)

# DJIA Data
DJIA <- fredr(series_id = 'DJIA')
DJIA_clean <- data.frame(
  date = DJIA$date,
  DJIA = DJIA$value
)

# US Dollar Index Data
DollarIndex <- fredr(series_id = 'DTWEXBGS')
DollarIndex_clean <- data.frame(
  date = DollarIndex$date,
  DollarIndex = DollarIndex$value
)

# Merge Data
data_combined <- reduce(
  list(SP500_clean, VIX_clean, PPIACO_clean, DJIA_clean, DollarIndex_clean),
  full_join,
  by = "date"
) %>%
  filter(complete.cases(.))

# Log Returns Calculation
log_returns <- data_combined %>%
  mutate(across(-date, ~ log(. / lag(.)))) %>%
  drop_na()
```

```
# Filter Date Range
log_returns <- log_returns %>%
  filter(date > as.Date('2019-09-01'))

# Summary Statistics
summary_stats <- log_returns %>%
  summarise(across(-date, list(mean = mean, sd = sd))) %>%
  pivot_longer(cols = everything(), names_to = c("Asset", "Metric"), names_sep = "_") %>%
  pivot_wider(names_from = Metric, values_from = value)
print(summary_stats)

# Fit Marginal Distributions
fit_t_distribution <- function(data) {
  MASS::fitdistr(
    x = data,
    densfun = 't',
    start = list(m = mean(data), s = sd(data), df = 2),
    lower = c(0, 0.001, 1)
  )
}

# Apply `fit_t_distribution` to Columns
params <- log_returns %>%
  dplyr::select(-date) %>%
  map(fit_t_distribution)
print(params)

# Create Pseudo Observations
pseudo_obs <- pobs(as.matrix(log_returns %>% dplyr::select(-date)))

# Fit Copula Model
vine_fit <- RVineStructureSelect(pseudo_obs, progress = TRUE)

# Simulate Data
simulations <- data.frame(RVineSim(N = 2560, RVM = vine_fit))

# Scale Simulations Back
scale_to_marginals <- function(column, param) {
  (qt(column, df = param$estimate["df"]) * param$estimate["s"] + param$estimate["m"])
}

scaled_simulations <- simulations %>%
  mutate(across(everything(), ~ scale_to_marginals(.x, params[[cur_column()]])))

# Add Grouping for 10-Day Returns
scaled_simulations <- scaled_simulations %>%
  mutate(Simulation = rep(1:(nrow(.) / 10), each = 10))

# Calculate 10-Day Returns
calculate_10_day_return <- function(initial_value, asset_values) {
  prod(1 + asset_values) - initial_value
}

returns_summary <- scaled_simulations %>%
  group_by(Simulation) %>%
  summarise(across(everything(), ~ calculate_10_day_return(initial_value = 1, .)))

# Export Results
write.csv(returns_summary, 'path_to_output_file.csv')

# Backtesting: Fit and Evaluate Alternative Copulas
t_fit <- tCopula(dim = ncol(pseudo_obs), dispstr = 'un')
normal_fit <- normalCopula(dim = ncol(pseudo_obs), dispstr = 'un')

fit_t <- fitCopula(t_fit, data = pseudo_obs)
fit_normal <- fitCopula(normal_fit, data = pseudo_obs)

# Extract LogLikelihood, AIC, and BIC from fitCopula objects directly
t_loglik <- logLik(fit_t)
t_aic <- AIC(fit_t)
t_bic <- BIC(fit_t)

normal_loglik <- logLik(fit_normal)
normal_aic <- AIC(fit_normal)
normal_bic <- BIC(fit_normal)

# Create Data Frame for Comparison
copula_metrics <- data.frame(
  Model = c("t-Copula", "Normal Copula"),
  LogLikelihood = c(t_loglik, normal_loglik),
  AIC = c(t_aic, normal_aic),
```

```
  BIC = c(t_bic, normal_bic)
)

# Print the metrics
print(copula_metrics)

# Visualizations
# Line Plots for Asset Returns
SP_Line <- ggplot(log_returns, aes(x = date, y = SP500)) +
  geom_line(color = "blue") +
  labs(title = "S&P 500 Returns", x = "Date", y = "Log Returns") +
  theme_minimal()

VIX_Line <- ggplot(log_returns, aes(x = date, y = VIX)) +
  geom_line(color = "red") +
  labs(title = "VIX Returns", x = "Date", y = "Log Returns") +
  theme_minimal()

PPIACO_Line <- ggplot(log_returns, aes(x = date, y = PPIACO)) +
  geom_line(color = "green") +
  labs(title = "PPIACO Returns", x = "Date", y = "Log Returns") +
  theme_minimal()

DJIA_Line <- ggplot(log_returns, aes(x = date, y = DJIA)) +
  geom_line(color = "purple") +
  labs(title = "DJIA Returns", x = "Date", y = "Log Returns") +
  theme_minimal()

DollarIndex_Line <- ggplot(log_returns, aes(x = date, y = DollarIndex)) +
  geom_line(color = "orange") +
  labs(title = "US Dollar Index Returns", x = "Date", y = "Log Returns") +
  theme_minimal()

# Combine Line Plots
Line_Plots <- ggpubr::ggarrange(SP_Line, VIX_Line, PPIACO_Line, DJIA_Line, DollarIndex_Line, ncol = 2, nrow = 3)
print(Line_Plots)

# Correlation Matrix Heatmap
CorrelationMatrix <- cor(log_returns %>% dplyr::select(-date), method = 'spearman')
print(CorrelationMatrix)

Corrplot_Data <- as.data.frame(as.table(CorrelationMatrix)) %>%
  rename(Asset1 = Var1, Asset2 = Var2, Correlation = Freq)

Corrplot <- ggplot(Corrplot_Data, aes(x = Asset1, y = Asset2, fill = Correlation)) +
  geom_tile() +
  scale_fill_gradient2(low = '#D02915', high = '#18874C', mid = '#F3E6F8', midpoint = 0,
                       name = 'Spearman Correlation') +
  labs(title = 'Correlation Matrix: Asset Returns',
       caption = 'Source: St. Louis Fed') +
  theme_minimal()

# Display the correlation heatmap
print(Corrplot)

# Validate Simulations
simulated_correlation <- cor(scaled_simulations, method = 'spearman')
print(simulated_correlation)

# Compare Actual vs Simulated Statistics
actual_stats <- log_returns %>%
  summarise(across(-date, list(mean = mean, sd = sd)))
sim_stats <- scaled_simulations %>%
  summarise(across(everything(), list(mean = mean, sd = sd)))
comparison <- rbind(
  Actual = unlist(actual_stats),
  Simulated = unlist(sim_stats)
)
print(comparison)

# Copula Simulations vs Actual Data (Comparisons)
plot_copula_comparison <- function(asset1, asset2, label1, label2) {
  real_plot <- ggplot(log_returns) +
    geom_point(aes_string(x = asset1, y = asset2), alpha = 0.5) +
    geom_smooth(aes_string(x = asset1, y = asset2), method = 'lm') +
    labs(title = paste(label1, "vs.", label2, "(Actual)"),
         x = paste(label1, "Returns"),
         y = paste(label2, "Returns")) +
    theme_minimal()

  sim_plot <- ggplot(scaled_simulations) +
    geom_point(aes_string(x = asset1, y = asset2), alpha = 0.5) +
```

```
    geom_smooth(aes_string(x = asset1, y = asset2), method = 'lm') +
    labs(title = paste(label1, "vs.", label2, "(Simulated)"),
         x = paste("Simulated", label1, "Returns"),
         y = paste("Simulated", label2, "Returns")) +
    theme_minimal()

  ggpubr::ggarrange(real_plot, sim_plot, ncol = 2)
}

# Generate and Print Comparisons for All 5 Asset Classes
comparisons <- list(
  SP_Vix_Comparison = plot_copula_comparison("VIX", "SP500", "VIX", "S&P 500"),
  SP_DollarIndex_Comparison = plot_copula_comparison("DollarIndex", "SP500", "US Dollar Index", "S&P 500"),
  PPIACO_DollarIndex_Comparison = plot_copula_comparison("DollarIndex", "PPIACO", "US Dollar Index", "PPIACO"),
  DJIA_PPIACO_Comparison = plot_copula_comparison("DJIA", "PPIACO", "DJIA", "PPIACO"),
  DJIA_Vix_Comparison = plot_copula_comparison("DJIA", "VIX", "DJIA", "VIX")
)

# Display All Comparisons
print(comparisons$SP_Vix_Comparison)
print(comparisons$SP_DollarIndex_Comparison)
print(comparisons$PPIACO_DollarIndex_Comparison)
print(comparisons$DJIA_PPIACO_Comparison)
print(comparisons$DJIA_Vix_Comparison)

# View the first few rows of the 10-day returns
print(returns_summary)
```